Supplementary Experimental Protocols and Results

Renyuan Wang[1], Yuhai Zhao[1,*], Jing Lu[2], Xingwei Wang[1]
[1]School of Computer Science and Engineering, Northeastern University, Shenyang, China
[2]College of Information, Liaoning University, Shenyang, China
wangry4@mails.neu.edu.cn, zhaoyuhai@mail.neu.edu.cn,
lujing@lnu.edu.cn, wangxw@mail.neu.edu.cn
[*]Corresponding author: Yuhai Zhao

TABLE I
Data preparation. Raw counts precede filtering and deduplication; prepared counts are unique coordinate pairs. Component trajectory controls use 2M/3M subsets; the fixed-query online comparison uses 3M for both datasets.

Dataset	Raw records	Prepared points
NE	123,593	118,620
CAS	196,902	124,514
TS	389,942	383,852
GeoLife	24,876,978	18,355,282
TDrive	17,662,984	11,274,655

This document provides reproducibility settings, complete results for the main comparisons, mechanism controls, scalability measurements and correctness proofs for DiSMC. Each control retains its workload, acquisition policy and timing scope; fixed-state replays and complete online queries are reported separately. The source release documents build, test and synthetic-smoke commands; historical large-scale timing records are not bundled.

I. Experimental Setup and Reproducibility

A. Settings, comparators and timing

We evaluate reference-answer preservation, E2E and scaling. NE/CAS/TS use the original ESMC coordinates. GeoLife and TDrive are Microsoft Research trajectories [1], [2]. Preprocessing retains valid WGS84 records in $[115, 118]^\circ$E and $[39, 42]^\circ$N, projects to EPSG:32650 (meters), rounds to three decimals, globally deduplicates coordinate pairs and subtracts each dataset's coordinate minima. Table I separates raw records from prepared unique points. Fixed-seed reservoir sampling produces shuffled nested subsets: GeoLife uses seed 20260925, TDrive 20260924. GeoLife 2M is the first 2M records of its frozen 3M sample; 1M subsets are prefixes of the corresponding 3M sample. Input hashes and preprocessing metadata accompany the artifacts. Mechanism defaults are $k = 2$, 200 owners, 40 cores and a 5% cap; queries and diagnostic caps are specified per control.

The cluster has five KVM workers, each exposing 12 vCPUs and 19.61–23.47GiB of guest memory. Spark 3.2.0 runs with Java 1.8.0_211. Distributed configurations request 40 executor cores, four cores and 4GB per executor, with a 4GB driver for component controls and an 8GB driver for complete online Planner and full-data queries. Scaling varies virtual cores. Locality waiting is disabled. The complete online scaling protocol is reported below; frozen-state controls retain their responsibilities and probe identities.

E2E includes the output and fingerprinting scope specified by each control. The fixed-state selector replay excludes capture and reconstruction; other replays retain their stated timing scopes. Repeated controls and both full-data cases report three trials. Online Planner, selector-replay and native calibration tables use means and sample standard deviations; component timing-range tables use medians and observed ranges. Main-text Engine transfer results use means, while supplementary transfer tables retain medians. For the fixed-query stage decomposition, per-method stage means sum to the corresponding mean E2E. Full-data phase profiles are separate: GeoLife reports one profiled execution alongside a three-run E2E mean, whereas TDrive reports phase records for all three trials. The main-paper unified-protocol table lists the acquisition controls; their percentage reductions are not additive components of the complete-query gain.

Distributed methods share prepared records and complete-output timing. DiSMC and Static-Greedy use a common squared-distance counting/export predicate; the supplied operator coverage path uses it as well. Legacy ESMC-based comparators retain two-decimal distance tests, whose floating-point boundary decisions need not be identical. System comparisons include planning and local execution differences; matched controls compare work signals within DiSMC. The assembled Spark operator comparator uses spatial filtering, geometric deduplication and complete aggregation, with spillable streaming coverage. System comparisons use 200 logical partitions, source-grid multiplier $\beta = 1$, radii 20/75 for GeoLife/TDrive, and $k = 2$. Grid/Hilbert/Z-order hold the local ESMC kernel fixed while changing spatial ownership and halo organization; the operator pipeline uses spillable coverage aggregation. Launcher arguments and required environment overrides are documented in the source-release README. Unit, Static and Sampled are matched-information scoring controls using Equation 1. Static-Greedy is a separate complete-planning comparator: it performs no online probes and has no completed-anchor or probe-witness transfer.

1) Hardware details: The heavy-query Planner comparisons use 200 owners, 40 executor cores, an 8GB driver and a 5% additional-input cap. Five KVM workers expose 12 vCPUs

TABLE II
Unified workload and acquisition registry. Paired values are GeoLife/TDrive; $q_E = (15, 15)$ and q_M denotes the dataset-specific medium location. All distributed rows request 40 cores unless cores are varied.

Control	Points	Query / radius	Cap
System	1M/3M	q_E, 20/75	5% (DiSMC)
Online queries	3M/3M	Four ranks, 100/300	5%
Input policy	2M/3M	q_M, r/r_0	5%
Calibration	2M/3M	q_M, 40/150	5%
Selector	2M/2M	q_M, 40/40	5%
Candidates	2M/3M	q_M, 40/150	5%
Probe reuse	Frozen inputs	q_E, fixed	Frozen plan
Count grid	Frozen inputs	q_E/q_M, fixed	Frozen plan
Core scaling	3M/3M	q_M, 60/200	5%
Full data	18.36M/11.27M	q_M, 20/200	10%

TABLE III
Independent correctness checks. Local checks verify returned counts; Spark checks verify complete members. Numbers are input–configuration checks.

Input	Local checks	Spark checks	Frontier errors	Count/member errors
Boundary	30	–	0	0
Random	60	–	0	0
NE	15	3	0	0
CAS	15	3	0	0
TS	15	3	0	0
GeoLife	15	3	0	0
TDrive	15	9	0	0
Total	165	21	0	0

each and 19.61–23.47GiB of guest memory. Four guests expose Intel Xeon E5-2420 CPUs at 1.90GHz; the fifth exposes a generic QEMU CPU model. Workers communicate over a private /24 IPv4 subnet using Virtio Ethernet adapters. Spark 3.2.0 runs with Java 1.8.0_211; ten executors each request four cores and 4GB. Dense boundary counting, counting-resolution multiplier two and locality waiting disabled are fixed. Acquisition considers up to 200 owners and sixteen tasks per alternative; subdivisions are selected without manual owner overrides.

B. Correctness validation

An independent exhaustive oracle implements the reference candidate construction, rounded predicates and dominance rules without calling DiSMC's indexes, pruning, counting cache or reducer. It enumerates single-point and eligible pair configurations, scans the complete input for coverage and computes the objective frontier. Inputs comprise frozen small subsets of the five datasets, six handcrafted boundary cases and twelve fixed-seed random cases. Table III reports zero frontier differences. Local checks vary ownership, full-input pieces, order and counting resolution; Spark checks include probes, completion/witness combinations and exact members. This validates the tested finite-candidate configurations. Canonical identity remains a separate tie-safe contract. The additional execution-comparison checks report zero frontier, representative, same-circle member and count differences in 30 recorded checks. These checks complement the independent oracle; the 2026-11-02 source release contains the relevant regression tests but does not bundle the historical run IDs or raw execution logs.

C. Correctness scope and artifact entry points

The mathematical SMCQ is continuous; ESMC supplies the finite single-/pair-point reduction. DiSMC's theorem concerns the resulting numerical reference, conditional on complete dependencies and safe pruning. It does not supply a new real-arithmetic reduction proof. Rounding the distance objective and using the documented coverage predicate define the implementation target. Constructor and boundary regressions test agreement on finite cases; testing does not establish a universal floating-point theorem.

One representative per non-dominated objective is an objective-equivalence answer: every reported distance–cardinality pair is realized by a valid reference circle with all its covered source records. It is not an exhaustive enumeration of every tied center or every tied member set. Applications requiring all tied groups need an extended output contract. Coordinate tie-breaking among survivors is weaker than globally canonical representative preservation.

The 2026-11-02 source release contains 117 production Java files, 15 test sources, build and launch scripts, a five-point synthetic fixture, a source manifest and a verification record. It requires JDK 8 and Spark 3.2.0 with Scala 2.12 jars. The release records a successful build, 14 selected regression tests and a local[2] five-point member-export smoke test in that environment. It excludes compiled JARs, real datasets, historical large-scale performance logs, table generators and superseded source trees. The corrected implementation shares one squared-distance predicate between reference coverage and member export; actual-center inclusion certificates guard anchor-count rejection, and a failed certificate leaves the candidate for counting. The integrated release has not rerun the paper's large-scale performance suite, so previously reported timings are not relabelled as measurements of this source release. Real-data corpora must be obtained separately.

D. Comparator definitions and provenance

1) Comparable decisions and contribution boundary: LocationSpark's query scheduler includes sampled local work, repartitioning, re-indexing and result merging; its plan-generation formulation constrains available partitions and its greedy algorithm accepts useful splits [4]. AQWA minimizes workload-dependent scanning cost through persistent partition adaptation, accounting for split overhead and partition-size constraints [5]. Table IV compares these stated objects and constraints, rather than claiming that prior systems lack cost-aware partitioning. The finite-family threshold guarantee in DiSMC concerns complete acquired alternatives, not the unrestricted repartitioning problem considered by LocationSpark. The SMCQ-specific contribution is the executable alternative representation and its coordinated use: closure preserves generation/counting dependencies, exact charges determine feasible combinations, and completion/visibility separation permits probe reuse. Full/Dependency, fixed-state selector/presentation and Engine-transfer controls test these respective effects. Their percentage gains are not additive, and the complete online comparison changes more than the final selector.

2) Operator pipeline and implementation scope: The spatial-operator comparator is an assembled SMCQ pipeline, not a claim of native LocationSpark, AQWA, GeoSpark or other platform performance. The supplied variants share eligible single/pair candidate construction with $\mathrm{ReferenceCandidateGeometry}$, exact geometry-key deduplication, complete coverage aggregation, local/global objective reduction and original-source member assembly. They persist core input, use x-sorted neighbor scanning with distance/ID filters, and batch local frontier reduction. Table V records the differences present in the source snapshots. The main-source variant is in $\mathrm{artifact/src/}$; the historical sorted-coverage variant is in $\mathrm{artifact/baselines/streaming_snapshot/}$. In contrast, the additional online Planner comparisons use the same partitioned member-output mode in both arms. Operator-versus-DiSMC system gains therefore include execution and materialization differences; the online and same-state controls provide the component evidence.

3) Measurement provenance: System comparisons use the queries, radii, partitions and matched resources stated in the main text. Aggregate historical measurements and version-bound run records are distinguished in the artifact; supplied source snapshots do not bind every historical timing to a specific executable.

The eight-query online comparison has 48 saved commands, build identifiers and trial logs. Static-Greedy is an internal strategy control sharing the Engine, dependency construction and cap; it disables probes and their transfers. The fixed-state selector control instead holds acquired information and task presentation fixed. Neither comparison claims a tuned-best native LocationSpark baseline or independent reproduction of all historical timings.

II. Fixed-Query Complete Online Results

A. Eight fixed queries

Both datasets have 3M points, four preselected locations, 40 cores, 200 owners and a 5% cap; GeoLife uses $r = 100$, TDrive $r = 300$. Each arm has three complete online trials in alternating serial order. All 48 runs succeed, all 24 paired objective frontiers agree, and all member-count checks pass. DiSMC is faster in every paired repetition.

The query coordinates were fixed before timing from a frozen population-ranked sample. Their labels do not denote density quantiles at the evaluated radii. Complete E2E includes acquisition, probing, planning, routing, formal execution, reduction and member output. Geometric mean speedups are 2.107× on GeoLife and 1.799× on TDrive. The query is the coverage unit; repetitions describe within-query variability. Component effects are evaluated by the separate controlled experiments.

III. Planning and Execution Mechanisms

A. Dependency inputs and matched cost signals

$$\begin{aligned} \widehat{w}^{\text{unit}}(A) &= |A|, \\ \widehat{w}^{\text{static}}(A) &= \sum_{a \in A} N_{c(a)}^2, \\ \widehat{w}^{\text{sampled}}(A) &= \alpha_o \sum_{a \in A} N_{c(a)}^2. \end{aligned} \tag{1}$$

Layout controls share Engine and probe policy; sampled owners may change with layout. Point-layout/no-split, work-layout/no-split and work-layout/full-plan E2E medians are 264.906/122.941/94.970s on GeoLife and 169.224/106.707/69.628s on TDrive. Work layout reduces E2E by 53.59%/36.94%; planning adds 22.75%/34.75%.

Joint responsibility–input planning. Full/Dependency independently select plans with common work signals, Engine and query settings. Dependency reduces median E2E by 5.19–14.84% on GeoLife and 8.02–13.96% on TDrive, with separated three-run ranges. Lower input charges admit more tasks: TDrive at $0.75r_0$ uses 6,295 instead of 19,735 added records and 203 instead of 201 tasks.

Fixed-responsibility contraction. Figure S1(a) reports the matched-responsibility assignment-count control: Full/Dependency additional assignments are 48,163/33,223 on GeoLife and 270,702/71,670 on TDrive, reductions of 31.0% and 73.5%, respectively. A distinct fixed-responsibility budget-feasibility control at a 5% additional-record cap is reported in Table IX and Figure S1(b). GeoLife Full/Dependency charges are 308,095/174,393 against a budget of 180,314; TDrive charges are 983,097/259,343 against 265,418. Dependency therefore reduces the additional assignment charge by 43.40%/73.62%, occupying 96.72%/97.71% of the respective allowances, while the corresponding Full-parent charges exceed the allowances. This fixed-responsibility comparison isolates input-construction charges; the over-budget Full-parent alternatives are assessed

TABLE IV
Published optimization objects versus DiSMC's finite physical-plan family. Constraints are summarized for the cited mechanisms, not all possible extensions.

Method	Decision object	Costs and constraints	Relation to this paper
LocationSpark	Repartition indexed spatial data and assign range/kNN queries	Local execution, reorganization/re-indexing and merging; available partition count	Related greedy scheduling principle; different decision objects and resource accounting
AQWA	Incrementally adapt persistent spatial partitions to data/query workload	Query overlap times partition population; split overhead and partition-size constraints	Related workload-aware splitting; different from per-query SMCQ initiation and dependency inputs
DiSMC	Choose an acquired responsibility–complete-input alternative per owner	Estimated private peak; exact additional owner–record charge under a shared cap	Conditional finite-family optimum; complete online utility evaluated separately

TABLE V
Verified source-level organization of the two supplied operator variants. This table does not assign a historical timing to a binary.

Stage	Supplied main-source variant	Supplied streaming snapshot
Candidate generation	Core/halo $\mathrm{cogroup}$; reference geometry	Same core/halo organization
Geometry deduplication	Geometry-key reduction	Geometry-key reduction
Coverage	Routed candidates $\mathrm{cogroup}$ with points	Sorted repartitioned join; stream candidate counts while buffering current-cell points
Objective reduction	Partition batches, then global reduction	Same reduction organization
Member assembly	Original-source scan; $\mathrm{collectAsMap}$ to driver	Original-source scan; $\mathrm{collectAsMap}$ to driver

for feasibility only, not as budget-feasible E2E executions. The 2026-11-10 control used GeoLife 2M/$r = 20$ and TDrive 3M/$r = 75$, both at $q = (15, 15)$, $k = 2$, 200 owners, and 40 cores. Each dataset has one complete Dependency run. Full-parent charges were recomputed for the same selected responsibilities from recorded parent-input counts, without executing the over-budget Full-parent plan. Charges are additional task-input assignments, not bytes or timings. The budget is $B = \lfloor 0.05R_0 \rfloor$, where R_0 sums unsplit owner inputs rather than distinct records $|P|$. Run commands, hashes, and responsibility files are preserved with the experiment records; these are separate from the three-run input-policy E2E control.

Sampled reduces matched-information E2E over Static by 13.80%/9.69% and maximum task duration by 36.31%/37.45%, with separated E2E ranges on both datasets. Unit and Static retain the same witnesses and completed work, so this gain arises from the work signal used for selection and presentation. Native Static/Sampled include their respective acquisition costs: GeoLife medians are 108.565/94.665s with ranges [107.439–109.696]/[91.948–95.924]; TDrive medians are 79.153/69.628s with [78.428–79.634]/[68.095–70.440]. The resulting 12.80%/12.03% reductions show that the planning benefit remains after online acquisition.

B. Matched calibration

C. Coordinated selection with common information

The main fixed-order selector table uses 2M points, $r = 40$, a 5% record cap, 200 owners and 40 cores on both GeoLife and TDrive, with their medium query locations. Joint and Greedy share candidates, costs, complete inputs, completion state, retained circles and frontier, and use logical owner/task presentation. Three replays per arm measure routing through member output; acquisition and reconstruction are excluded.

GeoLife Joint times are 68.155, 68.907 and 64.140 seconds; Greedy times are 94.929, 90.242 and 91.735. The corresponding means and sample SDs are $67.067 \pm 2.563/92.302 \pm 2.394$ seconds, with mean longest tasks 38.860/59.584 seconds. TDrive Joint times are 91.500, 95.433 and 92.796 seconds; Greedy times are 110.187, 113.963 and 117.151. Their means and sample SDs are $93.243 \pm 2.004/113.767 \pm 3.486$ seconds. Joint longest tasks are 42.644, 41.698 and 43.207 seconds; Greedy longest tasks are 49.159, 47.326 and 52.306, giving means 42.516/49.597 seconds. Mean replay reductions are 27.34%/18.04%. These are conditional selection comparisons, not complete online-strategy comparisons.

The reported 5% selector batch is recorded under $\mathrm{artifact/records/selector_5pct_author_batch_20261023}$; its timing scope and repetitions are given above.

D. Candidate decisions and selected-task prediction

State and selection. This diagnostic uses the archived fixed-order build (SHA-256 prefix $\mathrm{ca5e66e04b9a}$), GeoLife 2M/$r = 40$ and TDrive 3M/$r = 150$, their original medium queries, 200 owners, 40-core capture requests and captured 5% record caps. It is separate from the 3M large-radius online cohort. Owner selection precedes candidate timings: the original frozen estimates identify the heaviest eligible owner, then a distinct owner with maximum dependency-input overlap. The resulting IDs are GeoLife 154/45 and TDrive 72/194. TDrive 194 has no saved complete input snapshot; it is recorded as missing, without replacement. New captures retain all available candidate costs before validation, without feedback from validation timings. The first acquired alternative at each requested granularity is used; this is not exhaustive candidate search.

Execution and timing. All 36 formal candidate executions succeed, three per candidate. Each candidate runs in a separate

TABLE VI
Fixed-query complete E2E, seconds; all three trials.

Data	Query	DiSMC			Static-Greedy		
		Trial 1	Trial 2	Trial 3	Trial 1	Trial 2	Trial 3
GeoLife	q20	1458.253	1336.806	1465.698	3429.093	2799.024	2424.103
GeoLife	q40	1220.674	1333.912	1383.307	3215.764	3428.028	3054.038
GeoLife	q60	878.621	966.033	1155.820	2400.098	2507.480	2724.751
GeoLife	q80	1330.485	1086.892	1155.427	1601.604	1815.550	2121.760
TDrive	q20	513.656	521.541	500.724	866.055	847.271	865.887
TDrive	q40	396.264	389.503	440.391	845.766	824.872	797.370
TDrive	q60	453.836	370.592	444.224	826.483	874.742	867.179
TDrive	q80	517.583	477.225	560.230	803.697	808.305	768.371

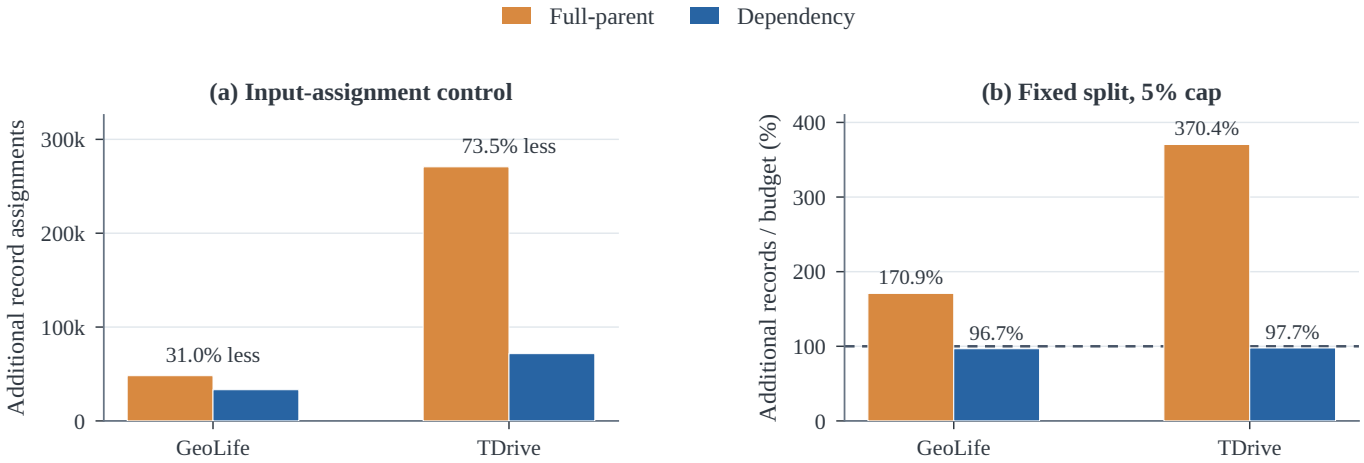


Fig. S1. Dependency-input contraction and budget feasibility in two distinct matched-responsibility controls. (a) Additional task-input record assignments, with reductions labeled. (b) Separate fixed-responsibility charge-to-budget ratios under a 5% additional-record cap; the dashed line marks 100% of the allowance. Over-cap Full-parent charges are feasibility comparisons, not measured E2E runs.

TABLE VII
Preselected coordinates for the fixed-query online evaluation.

Data	Query	q_x	q_y
GeoLife	q20	110607.167	92027.241
GeoLife	q40	111637.570	84896.430
GeoLife	q60	115704.130	107435.830
GeoLife	q80	122321.748	107853.175
TDrive	q20	136512.651	116175.856
TDrive	q40	112403.370	98369.878
TDrive	q60	122969.369	103183.266
TDrive	q80	120662.702	109883.629

JVM on the master host; tasks execute serially with independent private state and the same frozen external frontier. Complete responsibilities are executed without completion skipping, matching the scope of the predictions. Actual task time includes in-memory block construction, private-index preparation and the kernel; it excludes snapshot reading, network routing, Spark scheduling and member output. Peak is the maximum task time, and total work sums task times. These are controlled owner-level measurements, not cluster E2E. Planned and observed input/anchor populations match.

TABLE VIII
Full/Dependency input-planning control: three-run median E2E (seconds), fixed acquisition before calibration, common work signal and 5% cap.

Data	r/r_0	Full	Dep.	Reduction
GeoLife	0.75	76.937	67.976	11.65%
GeoLife	1	110.397	104.668	5.19%
GeoLife	1.5	226.947	193.276	14.84%
TDrive	0.75	64.548	59.374	8.02%
TDrive	1	79.650	72.788	8.62%
TDrive	1.5	200.398	172.427	13.96%

TABLE IX
Fixed-responsibility input-charge accounting, authors' 2026-11-10 rerun (5% additional-record cap). B is the allowed extra record assignments; Full/Dep. are charges on matched responsibilities. Dep./B is Dependency budget use; Saved is the relative charge reduction.

Data	B	Full	Dep.	Dep./B	Saved
GeoLife	180,314	308,095	174,393	96.72%	43.40%
TDrive	265,418	983,097	259,343	97.71%	73.62%

Candidate frontiers, merged with the common retained probe results, agree within each owner.

TABLE X
Matched-information controls: fixed acquisition, common probes, alternatives, input populations, Engine and cap. E2E shows median [min–max] over three runs; Max task is the median of each run's longest successful formal Spark task.

	E2E (s)		Max task (s)	
Weights	Geo	TDrive	Geo	TDrive
Unit	121.164 [116.986–122.088]	108.467 [108.089–110.205]	37.353	43.818
Static	109.698 [105.434–112.067]	77.100 [76.231–78.805]	37.289	29.235
Sampled	94.565 [94.493–95.896]	69.628 [68.095–70.440]	23.748	18.288

TABLE XI
Native calibration control: three online trials per arm. Complete E2E mean $\pm$ sample SD, seconds.

Data	Private-only	Paired-parent
GeoLife	107.799 ± 2.374	99.714 ± 2.152
TDrive	134.441 ± 2.955	127.280 ± 2.005

TABLE XII
Fixed owner/task-order selector control: 2M points, $r = 40$, 5% cap, 40 cores and 200 owners. Three-replay mean $\pm$ sample SD and mean longest task (s); acquisition excluded.

Data	Selector	Replay E2E	Longest
GeoLife	Joint	67.067 ± 2.563	38.860
GeoLife	Greedy	92.302 ± 2.394	59.584
TDrive	Joint	93.243 ± 2.004	42.516
TDrive	Greedy	113.767 ± 3.486	49.597

Calibration scope. GeoLife 154 has a two-task private guide; four/eight-task candidates test transfer. TDrive 72 has a four-task guide; two/eight-task candidates test transfer. The guide-granularity label does not independently certify the identity of every equivalent candidate. GeoLife 45 has no private-guide update in this capture and tests the initial/shared-state estimator. Thus only two owners directly test private-guide transfer. Absolute predictions show substantial bias; correctness of a decision is evaluated separately from absolute accuracy.

Decision result. For each owner, define peak regret as the mean actual peak of the predicted minimum-peak feasible candidate divided by the smallest mean actual peak among tested feasible candidates, minus one. The predicted choices are eight tasks for all three owners; all have zero observed regret. All tested alternatives fit the 5% cap with other owners unsplit. Feasible peak ordering agrees in these small, mostly monotone candidate sets. The main candidate-decision table presents the feasible sets, choices and best/next-best peaks; this full table retains all alternatives. This supports decision utility for the measured states, not unbiased times, difficult rank-reversal cases, globally optimal multi-owner allocation or minimum online E2E. The separate fixed-order selector and complete online controls evaluate broader strategy effects. Figure S7 summarizes the predicted and measured owner-level peaks across the tested task counts.

Artifacts. Complete per-run logs, candidate aggregates, manifests and analyzer are under $\mathrm{artifact/records/}$ in $\mathrm{candidate_validation_20261023}$. The artifact retains execution logs and the missing-input record; no failed candidate timings are excluded.

1) Selected-task prediction diagnostics: The diagnostic script joins the saved $\mathrm{micro_cost_predictions.csv}$ and $\mathrm{micro_tasks.csv}$ files in the current-results evidence archive by owner and task (the log field lane). Input populations must agree; keys must match uniquely. All 48 large-radius runs pass these checks, yielding 10,701 matched task records. The original 48-run join uses existing logs; the separate TDrive ranking diagnostic is described below.

For each run, Spearman correlation is the Pearson correlation of average ranks (ties receive average ranks). Top-tenth recall uses $k = \lceil 0.1m \rceil$ for m tasks, and divides the overlap between the k largest predictions and the k longest observed local task durations by k. Boundary ties use stable input order. The GeoLife diagnostic gives median correlation 0.590 and median recall 81.8%. The independent TDrive ranking diagnostic uses the same four query locations and three repetitions per query (12 runs), with unchanged 3M-point input, radius 300, 40-core resource request, budget, acquisition configuration and code version. It gives median correlation 0.571 and median recall 79.83%. The ranking diagnostic and the 48-run stage analysis are separate measurements under the same TDrive configuration. These are descriptive statistics over the fixed workload, not population confidence claims. Figure S2 summarizes the two dataset-specific ranking diagnostics.

The local-work sum W adds task-local wall durations; it is neither CPU time nor formal-stage wall time. Stage means in the main paper aggregate four queries and three repeats per method. The tail ratio is the maximum local task duration divided by formal-stage duration. Predictions concern full responsibilities, while formal execution skips completed anchors. Accordingly, this analysis tests ranking on executed alternatives, not absolute unbiased prediction or the costs of unselected alternatives. The tail recall supports prioritizing heavy tasks, but does not validate unselected alternatives or establish absolute prediction accuracy. Figure S3 visualizes the corresponding task-wall and tail-ratio summaries. Figure S6 reports the fixed-query aggregate stage contributions, using differences between 12-run method means rather than per-run samples.

The diagnostic evidence includes the analysis script, the per-run diagnostic table, and the joined per-task records. Paths are resolved from the manuscript workspace; source inputs are $\mathrm{artifact/current_results_20261019/large_radius_per_run.csv}$ and its $\mathrm{evidence.zip}$.

E. Counting-resolution control

Generation is fixed while the counting multiplier varies. GeoLife uses $q = (15, 15)$; TDrive uses the medium-density

TABLE XIII
All candidate measurements, seconds. Actual peak: mean ± sample SD over three repetitions; actual total: mean sum of task times. G/T denote GeoLife/TDrive and numbers are owner IDs. Additional records and feasibility use other owners unsplit: the cap is $B = \lfloor 0.05 R_0 \rfloor$. All twelve tested alternatives fit this cap.

Owner	Tasks	Pred. peak	Actual peak	Pred. total	Actual total	Extra records	Fits
G154	1	114.927	43.386 ± 0.627	114.927	43.386	0	yes
G154	2	67.474	27.435 ± 0.181	103.768	42.091	12,708	yes
G154	4	37.263	15.392 ± 0.328	103.792	42.217	42,240	yes
G154	8	17.971	9.084 ± 0.076	103.839	42.744	100,698	yes
G45	1	19.499	12.804 ± 0.159	19.499	12.804	0	yes
G45	2	10.877	7.822 ± 0.041	19.548	12.718	10,726	yes
G45	4	6.465	5.337 ± 0.034	19.656	12.680	34,675	yes
G45	8	3.285	3.197 ± 0.016	19.871	12.568	82,237	yes
T72	1	60.735	48.679 ± 0.525	60.735	48.679	0	yes
T72	2	37.394	29.865 ± 0.258	60.970	46.991	9,070	yes
T72	4	25.670	20.891 ± 0.284	60.989	48.274	32,603	yes
T72	8	9.918	15.450 ± 0.118	61.021	46.620	74,594	yes

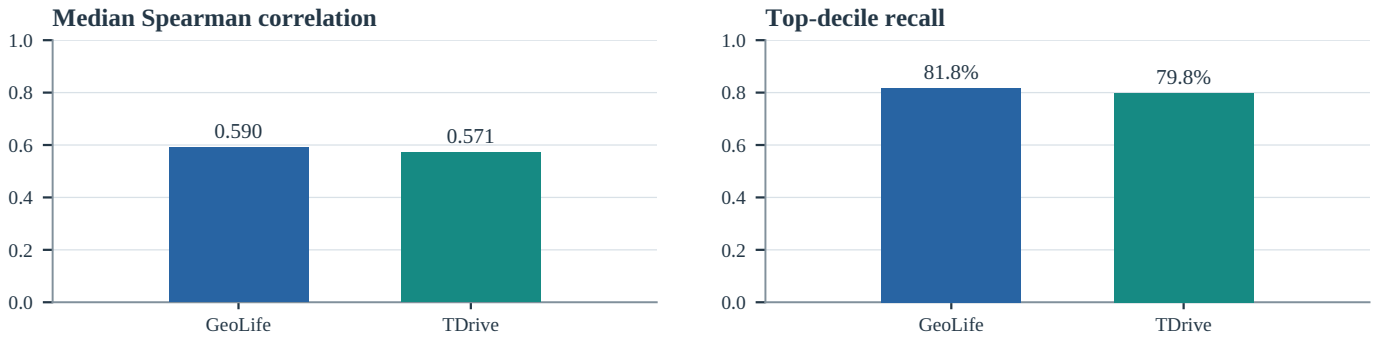


Fig. S2. Task-work ranking diagnostics. Bars show median Spearman correlation and top-decile recall. GeoLife summarizes the 48-run fixed-query diagnostic; TDrive uses a separate 12-run ranking diagnostic. These are descriptive workload summaries, not population confidence intervals.

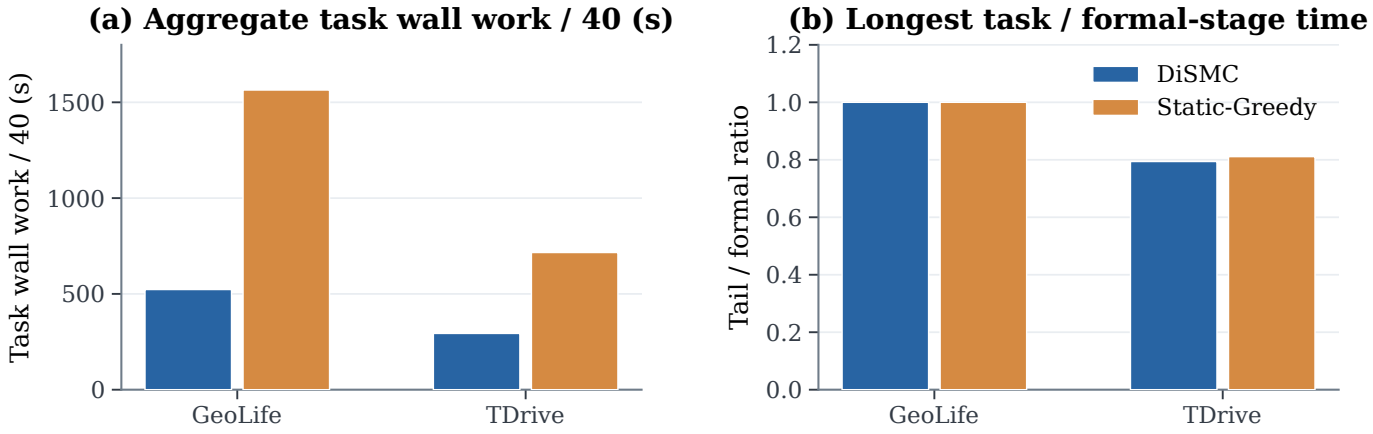


Fig. S3. Task-load diagnosis over the same 48 fixed-query runs. (a) Aggregate task wall duration divided by the 40-core request; $W/40$ is not CPU time. (b) Median longest-task/formal-stage ratio.

query $q = (116808.495, 114686.700)$. These timing results are separate from the completion/witness control. Each setting runs three times.

Factor 2 has the lowest median E2E on both datasets, reducing time by 4.09%/13.69% relative to factor 1, with separated observed ranges. Both finer and coarser settings are slower. Finer grids reduce point checks but increase E2E; savings do not offset other indexing and bucket-processing work. This control measures combined costs. Factor 2 is the experimental default. Figure S8 shows the corresponding E2E and task-summed formal-check results.

TABLE XIV
Stage and task diagnosis of the same 48 fixed-query runs. Plan, Formal, E2E and $W/40$ are means over 12 runs per row (s). Tail/Formal is the median longest-task/formal-stage ratio. W sums task wall durations, not CPU time.

Data	Method	Plan	Formal	E2E	$W/40$	Tail/ Formal
GeoLife	DiSMC	140.089	1052.300	1230.994	522.882	1.000
	Greedy	13.056	2571.570	2626.774	1564.133	1.000
TDrive	DiSMC	51.842	382.102	465.481	293.731	0.794
	Greedy	11.420	787.580	833.000	716.039	0.811

TABLE XV
Counting resolution: three-run median E2E [min–max] in seconds. $h_c = \lambda \min(h_s, r/4)$; generation is fixed. Checks sum formal generation and counting operations across tasks.

λ	E2E (s) Geo	E2E (s) TDrive	Checks (10^9) Geo	Checks (10^9) TDrive
0.5	132.414 [131.847–138.503]	146.425 [145.810–152.741]	23.548	20.502
1	99.080 [98.403–100.266]	84.331 [84.048–86.183]	28.278	21.680
2	95.029 [94.325–95.582]	72.788 [71.263–73.841]	70.503	54.306
4	115.549 [115.311–118.027]	86.713 [84.963–91.337]	176.027	164.371

TABLE XVI
Completion reuse (R) and witness transfer (W): three-run median E2E [min–max] in seconds. All four configurations use identical responsibilities, inputs and probe IDs; only the two transfer switches vary. Checks are task-summed formal operations, unchanged across recorded repetitions.

R	W	E2E (s) Geo	E2E (s) TDrive	Checks (10^9) Geo	Checks (10^9) TDrive
0	0	103.257 [102.852–105.827]	82.582 [80.231–84.824]	140.804	112.340
0	1	96.375 [95.643–97.048]	73.422 [72.663–74.826]	71.026	33.446
1	0	101.500 [99.446–101.968]	78.011 [77.057–78.542]	140.121	112.171
1	1	93.492 [92.793–94.582]	69.859 [68.628–71.142]	70.503	33.332

F. Completion and witness reuse

The reuse control fixes responsibilities, complete inputs and probe IDs, and varies only completion reuse and witness transfer. Both datasets use $q = (15, 15)$; each configuration runs three times.

Main-text mean comparisons show witness E2E reductions of 7.33%/10.79% and additional completion reductions of 2.84%/5.11% relative to witness-only execution. Formal checks fall by 49.56%/70.23% with witnesses. The tables report medians and observed ranges. Each repetition identifies the same 192 completed anchors. With completion reuse enabled, 1,999,808/2,999,808 anchors remain for formal initiation; with reuse disabled, all responsibilities are initiated again. Dependency visibility is preserved in both cases. This 192-anchor controlled set is distinct from the 832 completed anchors in the separate captured-state accounting in Table XVII; the two counts describe different experimental states. On GeoLife, the counting-resolution setting $\lambda = 2$ and the fully enabled reuse setting describe the same Engine configuration, explaining their identical operation count. These matching counts are not additional independent evidence. TDrive uses different query locations for the two controls, as specified above.

TABLE XVII
Actual completion accounting in the captured query states. Formal anchors exclude completed initiation but retain dependency visibility.

Data	Source anchors	Completed	Formal
GeoLife	2,000,000	832	1,999,168
TDrive	3,000,000	832	2,999,168

G. Completion accounting

H. Probe-cost estimation and transfer

This section specifies the existing estimator rather than introducing a new model. Cost estimates change candidate scores, not task responsibilities, complete dependencies or record charges. The directly measured private guide and its matched parent replay remain distinct. The artifact functions $\mathrm{SpatialProbeCostModel.reweight}$ and $\mathrm{LaneProbeRefinement.update}$ implement the following two stages.

1) Initial spatial-residual calibration: For an owner with observations, let $x_i = \max(0, K_i - I_i)$ be recurring measured work and let $v_i > 0$ be the cell-work proxy for observed anchor i. Regroup anchors by the alternative's task j, giving observation count m_j, $X_j = \sum_{i \in j} x_i$ and $V_j = \sum_{i \in j} v_i$. If the overall sums X, V and the task sum V_j are positive, the relative residual is $r_j = (X_j/V_j)/(X/V)$. For $m_j \geq 4$ and $V_j > 0$, use

$$c_j = \frac{m_j \, \min(4, \max(0.25, r_j)) + 4}{m_j + 4}; \tag{2}$$

otherwise $c_j = 1$. With incoming task estimate w_j^0, set $z_j = w_j^0 c_j$. The recurring total is $R = \max(0, \sum_j w_j^0 - F_o)$, where F_o sums measured owner preparation, setup and dense-boundary construction. For $Z = \sum_j z_j > 0$, set

$$\widehat{w}_j = R z_j / Z + F_o \, |V_{s,j}| / \max(1, |V_o|). \tag{3}$$

An owner without observations, nonpositive overall sums, or missing/nonpositive cell metadata retains its incoming alternative; the metadata branch also rejects nonfinite values. This stage uses clipped residuals and a four-sample prior, not the variance-dependent confidence below. Its owner-total calibration scales unmeasured-owner proxies by the median observed estimate/proxy ratio and uses measured estimates for observed owners.

2) Fresh private-guide observations and other layouts: Fix a measured guide g. Its task ℓ has $a_{g,\ell}$ anchors and $t_{g,\ell}$ fresh observations. Each observed anchor i receives $\omega_i = a_{g,\ell(i)}/t_{g,\ell(i)}$, with $x_i = \max(0, K_i - I_i)$. The owner recurring estimate is $R = \sum_i \omega_i x_i$. For another candidate layout s,

regroup these same observations by its own responsibility rule. In task j, let m_j be the observation count, $a_{s,j}$ the actual anchor population, and

$$W_j = \sum_{i \in j} \omega_i, \qquad n_j^{\text{eff}} = W_j^2 / \sum_{i \in j} \omega_i^2. \tag{4}$$

The effective count is zero when $W_j = 0$. For positive W_j, define $\mu_j = \sum \omega_i x_i / W_j$ and

$$v_j = \max\left(0, \frac{\sum_{i \in j} \omega_i x_i^2}{W_j \mu_j^2} - 1\right), \tag{5}$$

with $v_j = 0$ when $\mu_j = 0$. Here R is specific to the measured guide, not an invariant true owner cost. All costs use a common time unit. Each target candidate s maintains its own task prior weights $w^0_{o,s,j}$: its partition rule is applied to the complete anchor responsibilities, and cell-based workload contributions are aggregated for each resulting task, with the existing owner-level calibration. Thus, the target priors are not obtained by splitting the measured guide's task-level totals. Let $\mathcal{J}_{o,s}$ be the tasks of target alternative s. The time-valued prior allocation is $q_{o,s,j} = R w^0_{o,s,j} / \sum_{h \in \mathcal{J}_{o,s}} w^0_{o,s,h}$ (zero if the denominator is zero). This normalization uses the target alternative's own task priors, not base/guide-task indices or priors from other owners. For the fixed target below, abbreviate $q_{o,s,j}$ as q_j. If $W_j > 0$, $m_j \geq \min(4, a_{s,j})$ and $n_j^{\text{eff}} + 10^{-9} \geq \min(4, a_{s,j})$, use

$$\alpha_j = \frac{n_j^{\text{eff}}}{n_j^{\text{eff}} + 4v_j}, \qquad z_j = \alpha_j \mu_j a_{s,j} + (1 - \alpha_j) q_j. \tag{6}$$

Otherwise set $z_j = q_j$. For $Z = \sum_j z_j > 0$, rescale $z_j \leftarrow R z_j / Z$; when $Z = 0$, retain these zero values. Then allocate fixed setup using exact routed populations:

$$\widehat{w}_{s,j} = z_j + F_o \, |V_{s,j}| / \max(1, |V_o|). \tag{7}$$

For a private guide with measured task fixed costs $F_{g,j}$, the owner-equivalent fixed cost is $F_o = (\sum_j F_{g,j}) |V_o| / \sum_j |V_{g,j}|$. Crucially, the directly measured private guide bypasses the transfer and normalization: its task estimate is $\sum_{i \in j} \omega_i x_i + F_{g,j}$, as in the main guide equation. The matched full-parent estimate is calibrated independently. In the nonprivate observation branch, F_o is the measured preparation, setup, dense-boundary and nonnegative unassigned-kernel overhead retained by the implementation.

The prior q_j has units of time. The coefficient 4 and the minimum-count rule are fixed shrinkage heuristics, not a statistical confidence guarantee. For unmeasured layouts, normalization assumes the guide recurring total transfers while its distribution changes. Exact input populations change the fixed term, but layout-dependent changes in recurring counting, indexing and pruning are not fully modeled. Owners without new measurements retain the supplied fallback estimates. Observation identities must be fresh, distinct and consistent with requested guide quotas; private guide input populations must match the exact planned dependencies. The estimator retains the candidate family during an update. These are implemented transfer and validation rules, not guarantees of unbiased prediction or minimum online E2E.

I. Acquisition limits and guarded refinement

1) Bounded alternatives and information access: Probing starts when raw work exceeds ideal slot load or heavy-first scheduling exposes a finite-wave tail beyond summation roundoff. Up to 32 owners supply at most 24 initial anchors each. Calibrated work refreshes acquisition to reconsider owners excluded by raw scores.

The candidate family retains point-coordinate subdivisions, with depths two, four, eight and sixteen. Owner coverage and acquisition are bounded separately from probing. For N source points and available driver memory M, the coordinate allowance is

$$C(N, M) = \min\left(\left\lfloor \frac{0.15M}{2048} \right\rfloor, \ \max(50000, \lceil 0.01N \rceil)\right). \tag{8}$$

The bytes-per-anchor term is configured acquisition accounting, rather than measured heap use. A bounded spatial sample supplies cut coordinates when complete anchor acquisition would exceed the allowance. Exact task populations and conservative dependency inputs are evaluated against the full source; sampled cuts do not authorize sampled counts. At most 200 owners enter acquisition in the reported configurations. Additional-owner discovery coverage is an explicit parameter, c_{cov}, and is reported per experiment.

Private calibration uses at most 256 fresh anchors, with four to eight observations per nonempty guide stratum. Controls and full-data TDrive use at most four privately calibrated owners; full-data GeoLife uses eight. Each round measures one owner and reselects. Full-data GeoLife alternates standard rounds with coverage of eligible unsplit owners, falling back when none fits the observation allowance. No-split remains available at zero charge.

2) Extrapolating unmeasured layouts: For other layouts, regroup observations by responsibility with guide-stratum weights $\omega_i = a_{g,j}/t_{g,j}$. Blend observed and prior recurring work: $z_j = \alpha_j \mu_j a_j + (1 - \alpha_j) q_j$, where a_j is task population, μ_j its weighted mean, q_j its time-valued prior allocation from the measured guide recurring total R, and $\alpha_j = n_j^{\text{eff}} / (n_j^{\text{eff}} + 4v_j)$. Here $n_j^{\text{eff}} = (\sum \omega_i)^2 / \sum \omega_i^2$ and v_j is weighted squared coefficient of variation (zero for zero mean). Insufficient observed/effective counts use q_j. Normalize positive total z to R; allocate fixed cost by exact input size. Unmeasured owners retain fallback estimates. The measured guide uses the direct private-guide estimator in main-text Section III-D. In updatePaired, the parent replay independently updates ownerWork (the no-split cost), while private observations update population candidates; no parent/private multiplicative ratio is applied. The supplementary specification supplies count thresholds, zero-denominator rules and initial spatial-residual calibration.

3) Tail-refinement settings: **Guarded tail refinement.** Only privately measured owners with current peak at least half the primary peak are eligible. A deeper layout must

TABLE XVIII
Complete-system E2E at the external query. Three-run median E2E (s).

Method	GeoLife 1M	TDrive 3M
Grid	948.813	5906.491
Hilbert	852.340	4961.850
Z-order	1950.458	10990.405
Spatial operators	737.120	3990.405
DiSMC	43.433	69.628

TABLE XIX
Position coverage on 1M points per dataset: Three-run median E2E.

Data	Position	DiSMC (s)	Operator (s)	Ratio
GeoLife	Sparse	42.911	827.812	19.29
	Medium	42.227	715.550	16.95
	Dense	44.299	657.252	14.84
TDrive	Sparse	33.010	529.916	16.05
	Medium	33.475	533.129	15.93
	Dense	33.197	518.655	15.62

reduce its owner peak by at least 25%, respect the record cap and preserve the primary peak. Predicted heavy-first completion time and the sum of predicted task work must stay within 1% and 0.1% of the primary plan, respectively. The pass repeatedly selects the largest eligible owner peak, then its cheapest qualifying layout. It preserves estimated peak optimality, without guaranteeing minimum E2E or replication charge.

J. Counting implementation and acquisition complexity

For C nonempty cells and h neighbor offsets, statistics acquisition scans source records, cell ordering costs $O(C \log C)$ and neighborhood processing visits $O(Ch)$ relationships. Exact input-population scans depend on overlap and the candidate family; all acquisition and selection enter E2E.

Generation and counting use independent resolutions. With source width h_s, reference width $h_0 = \min(h_s, r/4)$ and positive multiplier λ, counting uses $h_c = \lambda h_0$, falling back to h_s if unsafe to represent. The default $\lambda = 2$ leaves generation fixed. For center z and bucket box Q, conservative $d_{\min}(z, Q)$ and $d_{\max}(z, Q)$ bounds certify disjointness or full coverage under the same monotone squared-distance predicate as individual points; uncertain buckets use exact tests. The visibility allowance τ does not enlarge counting acceptance. Dense boundary buckets use private refinement with leaf capacity eight and maximum depth eight, trading fewer point tests for construction and traversal.

IV. System Performance, Scalability and Full-Data Execution

A. Distributed system comparisons

Single-JVM ESMC references on NE/CAS/TS use one-machine resources; the distributed system comparisons below hold their resource requests fixed.

Table XVIII gives 16.97×/57.31× speedups over the GeoLife/TDrive operator pipeline. These compare complete execution organizations; the subsequent controls attribute improvements within DiSMC to its Planner and Engine.

A frozen population sample selects the 10th/50th/90th percentile neighborhoods without latency information: populations are 3/29/285 on GeoLife and 10/72/189 on TDrive. All six ratios in Table XIX exceed 14× with input and resources fixed within each dataset. These are query-position strata. Figure S4 visualizes the corresponding paired E2E results.

For Table XIX, GeoLife sparse/medium/dense query coordinates are $(121585.662, 102389.276)$, $(122287.250, 105282.863)$ and $(114693.645, 108474.258)$; TDrive coordinates are $(120924.613, 114579.245)$, $(106516.154, 134723.013)$ and $(126175.926, 103115.703)$. Radii are 20/75 and both inputs contain 1M points. These middle-stratum coordinates differ from the 3M Planner queries listed in the cross-location protocol.

TABLE XX
Single-machine versus distributed size sweep. Completed times are three-run medians in seconds. Timeout ratios are lower bounds computed from the 7200-second limit.

Data	Size	ESMC	DiSMC	Speedup
GeoLife	100K	22.261	22.969	0.969
GeoLife	200K	115.590	24.801	4.661
GeoLife	400K	1271.005	28.901	43.978
GeoLife	600K	Timeout	33.927	$\geq$ 212.220
GeoLife	800K	Timeout	38.153	$\geq$ 188.714
GeoLife	1M	Timeout	43.433	$\geq$ 165.773
TDrive	100K	20.205	20.194	1.001
TDrive	200K	103.165	21.586	4.779
TDrive	400K	862.560	25.750	33.497
TDrive	600K	4933.331	27.915	176.727
TDrive	800K	Timeout	27.605	$\geq$ 260.822
TDrive	1M	Timeout	31.695	$\geq$ 227.165
Porto	100K	15.346	20.019	0.767
Porto	200K	79.861	20.723	3.854
Porto	400K	1026.332	22.115	46.409
Porto	600K	5254.248	23.084	227.614
Porto	800K	Timeout	23.887	$\geq$ 301.419
Porto	1M	Timeout	25.323	$\geq$ 284.327

B. Single-machine size series

Overlapping DiSMC timings in the single-thread comparison, system comparison and input-size sweep use the same reported system/size-series values; no new paired run is implied. E1 uses single-threaded ESMC, a 4GB JVM heap, $q = (15, 15)$, $k = 2$ and a 7200-second timeout. Both methods deduplicate coordinates; three-run medians and censored ratios follow the main-text protocol.

Porto Taxi trajectories from the ECML/PKDD 2015 challenge [3] are expanded, filtered to $[-9.5, -7.5]^\circ$ longitude and $[40.5, 42.5]^\circ$ latitude, and projected to EPSG:32629, yielding 17,565,472 points. Porto coordinates are rounded to three decimals, globally deduplicated and translated by subtracting coordinate minima. Seed 20260924 generates shuffled nested reservoir samples; size-sweep inputs are fixed prefixes. Metadata records source and prepared hashes.

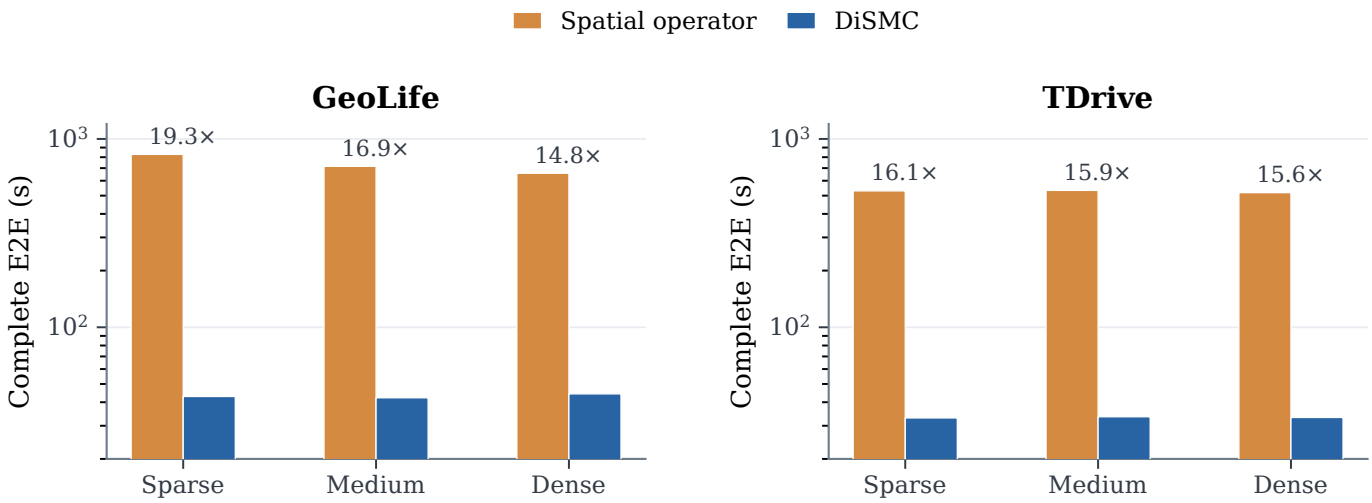


Fig. S4. Query-position robustness on 1M points per dataset. Three-run median complete E2E for sparse, medium and dense population strata; the assembled spatial-operator pipeline and DiSMC are compared within each dataset. Ratios are shown above paired bars.

TABLE XXI
Input-size E2E (s), fixed 40-core request and external query. Each entry reports three-run median E2E.

GeoLife			TDrive		
Size	DiSMC	Operator	Size	DiSMC	Hilbert
100K	22.969	35.962	400K	25.750	78.496
200K	24.801	51.942	600K	27.915	107.295
400K	28.901	116.851	1M	31.695	210.373
600K	33.927	235.954	2M	48.528	1204.983
800K	38.153	425.663	3M	69.628	4961.850

TABLE XXII
Complete online E2E, mean ± sample SD in seconds, and speedup relative to the same dataset's 8-core mean.

Cores	GeoLife	Speedup	TDrive	Speedup
8	707.833 ± 13.243	1.00	673.369 ± 2.379	1.00
16	373.965 ± 2.898	1.89	357.823 ± 19.090	1.88
24	297.917 ± 7.860	2.38	281.686 ± 5.043	2.39
32	245.910 ± 8.417	2.88	246.955 ± 10.743	2.73
40	212.286 ± 5.461	3.33	195.531 ± 1.035	3.44

C. Input and executor-core scalability

Table XXI holds one comparator fixed per dataset and shows increasing absolute gaps. A separate medium-position GeoLife series reaches 5M points (792.406s); its query differs and it is not pooled with this curve.

Each execution repeats online planning and includes complete output. Input tasks are coalesced to the core request; logical ownership remains 200. Private calibration uses four owners and 256 fresh anchors, and additional discovery coverage uses two anchors per owner. The observed 8-to-40-core E2E speedups are 3.33×/3.44×. The TDrive 40-core endpoint uses three measurements (196.455, 194.412 and 195.726 seconds), with mean 195.531 ± 1.035 seconds.

TABLE XXIII
GeoLife full data, 40 cores and a 10% cap. Mean ± sample SD uses three E2E trials; phase, plan and task metrics describe one profiled execution. Times are seconds.

Metric	DiSMC
Prepared points	18,355,282
Physical tasks	259
Additional records / cap	3,090,898 / 3,324,800
Reading and runtime preparation	21.547
Planning, including all probes	589.388
Routing and input preparation	31.346
Formal execution	2101.554
Global reduction	0.661
Complete member materialization	9.522
Remaining output/framework time	4.188
Longest task	1991.219
Profiled execution E2E	2758.206
Three-run mean E2E	2778.071 ± 106.580

D. Query and ownership sensitivity

Figure S5 varies one parameter at a time with acquisition configuration fixed and medium queries (GeoLife 2M/TDrive 3M), using three runs including online costs. Defaults are $k = 2$, 200 owners and 40 cores. Radius factors are 0.5/0.75/1/1.5/2; owner counts are 50/100/200/400/800.

Larger radii increase E2E; higher k decreases it. Owner count is non-monotone, with lowest observed times at 400/200 on GeoLife/TDrive. The cap remains 5% but its absolute value changes with R_0.

E. GeoLife full-data execution

The full-data case uses GeoLife 18,355,282 points, $q = (124414.676, 106577.117)$, $r = 20$, $k = 2$, 200 initial owners and 40 cores. The additional-record cap is 10%; $R_0 = 33,248,000$ gives an allowance of 3,324,800 records. The selected plan charges 3,090,898 records and has 259

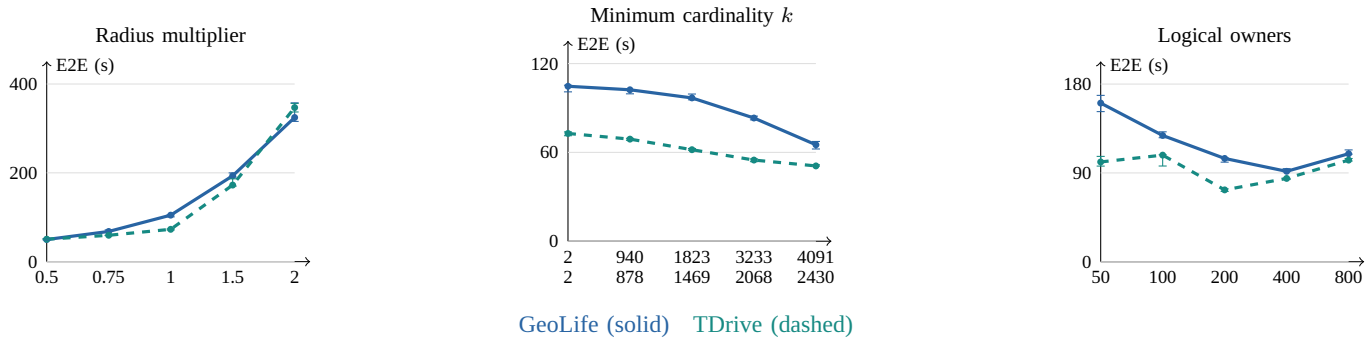


Fig. S5. DiSMC parameter sensitivity with fixed acquisition configuration; points show three-run median E2E and whiskers the observed minimum–maximum range. Minimum-cardinality settings are $\{2, 940, 1823, 3233, 4091\}$ on GeoLife and $\{2, 878, 1469, 2068, 2430\}$ on TDrive; upper/lower tick labels give GeoLife/TDrive values. Each panel varies one parameter.

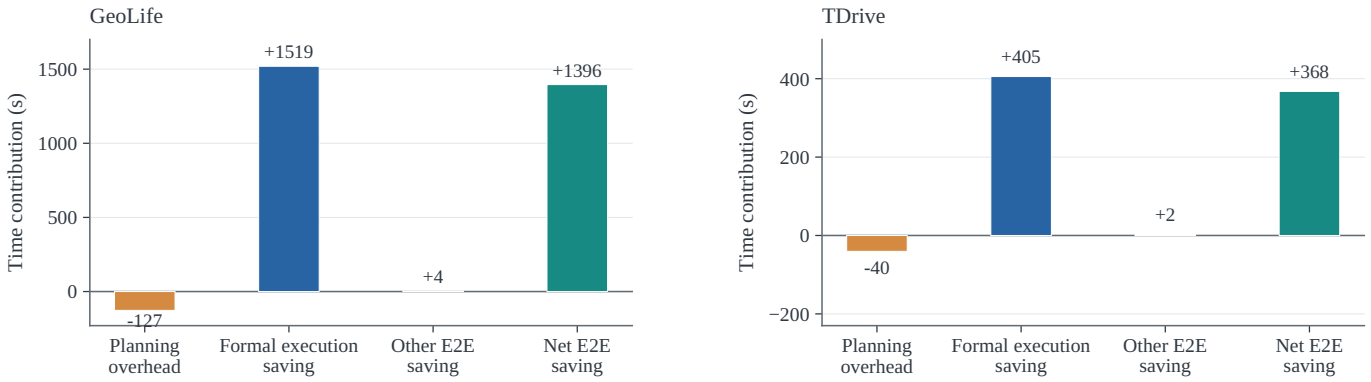


Fig. S6. Stage contribution to complete online E2E on the same fixed-query cohort. Bars show differences between the 12-run means per dataset and method: additional planning cost is negative, while formal-execution and net E2E savings are positive. Other E2E savings are 3.543 s/2.463 s on GeoLife/TDrive. No per-run standard deviation is inferred from the aggregated stage means.

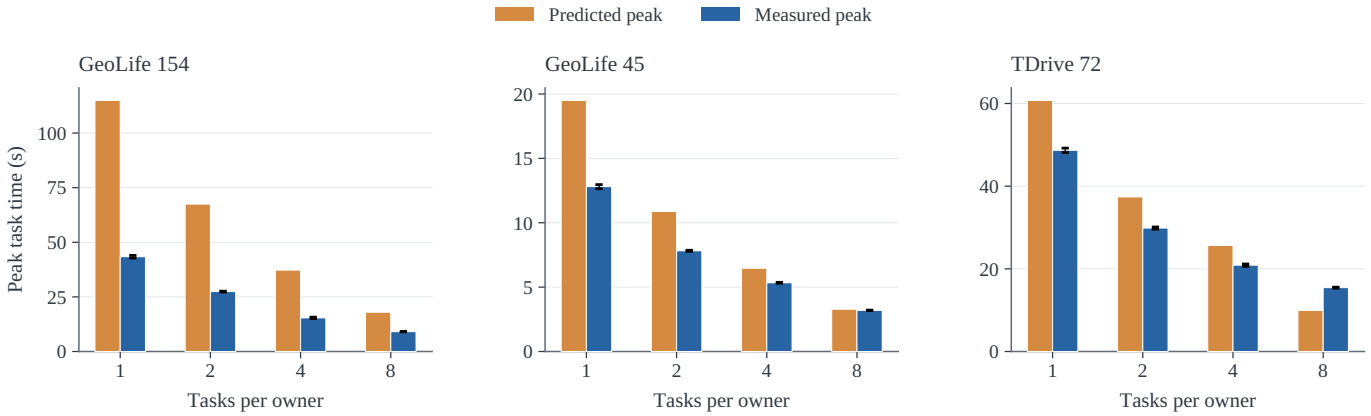


Fig. S7. Owner-level candidate peak diagnostic for three selected owners. Bars compare predicted peak with measured peak (mean ± sample SD over three serial repetitions) across 1, 2, 4 and 8 tasks per owner. These are controlled owner-level measurements, not cluster E2E or a general prediction-accuracy estimate; the predictions show substantial absolute bias. Each panel uses an independent vertical-axis range.

physical tasks. Initial discovery retains the 32-owner bound, additional coverage uses eight anchors per owner, and private calibration uses at most eight owners and 256 fresh anchors. Alternating unsplit-owner coverage and guarded secondary refinement are enabled. The executable hash begins 6b0759cf; the complete hash, configuration and phase record accompany the full-data JSON and command artifacts.

Full-data E2E times are 2758.206, 2682.821 and 2893.186 seconds: mean 2778.071 ± 106.580 seconds (sample SD). Phase and task records profile the first execution, whose longest task takes 1991.219 seconds. Its circle metadata and exported member population match the reference; this

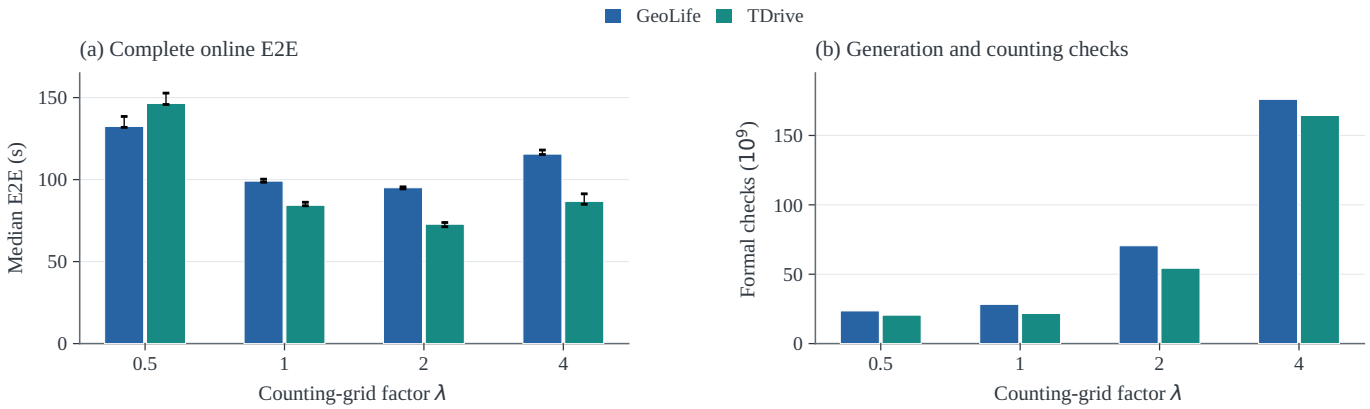


Fig. S8. Counting-grid resolution with candidate generation fixed. (a) Three-run median complete online E2E with observed min–max whiskers. (b) Task-summed formal generation and counting checks. $\lambda = 2$ gives the lowest tested E2E on both datasets; this control uses a different query location from the completion/witness reuse experiment.

TABLE XXIV
Longest tasks in the complete GeoLife run. Prediction and actual duration are seconds; pieces is the owner's total task count.

Owner	Task	Pieces	Predicted	Actual
96	1	2	1760.940	1991.219
99	1	2	1789.330	1796.919
100	0	2	1784.141	1641.178
99	0	2	1334.474	1621.014
95	0	2	1432.358	1519.054
45	0	4	1250.035	1466.095
44	0	2	1728.953	1456.615
39	2	4	1133.855	1384.706

TABLE XXV
TDrive full-data trials, $r = 200$, 10% cap. Seconds and record counts.

Trial	E2E	Longest	Tasks	Added records
1	2990.630	1743.790	301	3,531,240
2	3011.678	1532.159	299	3,526,185
3	2871.850	1223.390	304	3,486,610

TABLE XXVI
TDrive full-data stage times, seconds. Remainder closes E2E.

Stage	Trial 1	Trial 2	Trial 3
Read/preparation	17.897	18.734	18.622
Planning	238.305	228.502	205.580
Routing/preparation	21.201	22.043	22.238
Formal execution	2686.086	2715.023	2598.445
Global reduction	0.955	0.983	0.905
Member materialization	21.470	21.427	22.101
Remaining output/framework	4.716	4.966	3.959

TABLE XXVII
Original ESMC datasets: raw record counts before coordinate deduplication; three-run median E2E in seconds. Prepared sizes appear in the supplement.

Data	Raw records	ESMC	DiSMC	Speedup
NE	123,593	23.170	21.823	1.062×
CAS	196,902	57.100	20.522	2.782×
TS	389,942	2604.312	23.375	111.414×

check does not independently compare every member identity. Small-input member tests are reported separately. Table XXIV lists its longest tasks.

F. TDrive full-data execution

Three serial trials process all 11,274,655 prepared points at $q = (116808.495, 114686.700)$, $r = 200$, $k = 2$, 200 owners, 40 executor cores and an 8GB driver. The 10% cap allows 3,541,312 additional records. Coverage uses eight anchors per owner; private refinement uses up to four owners and 256 fresh anchors, with alternative depths up to sixteen tasks. The build SHA256 is recorded with actual commands and logs in the current-results artifact. E2E is 2958.053 ± 75.392 seconds (mean $\pm$ sample SD). All trials succeed, their objective frontiers agree, and each exports 366,242,992 circle-member records in 200 parts with valid aggregated member counts. Full member-set equality is not inferred from count agreement.

Figure S9 visualizes these same three-run median E2E values on a logarithmic axis; the table retains the exact timings.

V. Full Planner and Private Engine Algorithms

A. Budgeted joint selection (complete primary sweep)

B. Private Engine

The following listing is identical to the former detailed main-text Engine algorithm; the main text retains the execution contracts and correctness result.

VI. Correctness and Numerical Proofs

A. Numerical Analysis of Two-Level Dependency Inputs

This derivation concerns input routing and trimming. It does not replace the separate admissibility conditions for local pruning and probe transfer. Interpret each stored input as its exact binary real value. Assume source coordinates in $[0, M]^2$,

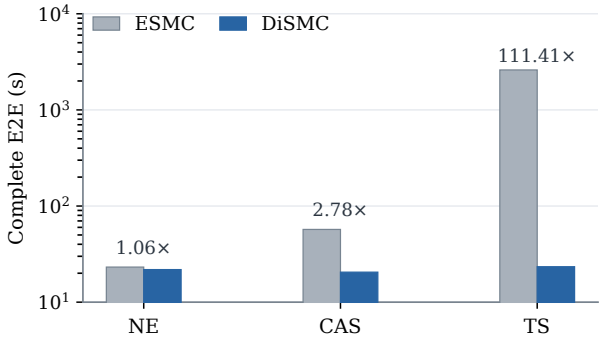


Fig. S9. Original ESMC benchmark inputs: three-run median complete E2E (log scale). Speedup annotations compare ESMC with DiSMC within each dataset. The two implementations differ in execution resources and pipeline semantics; these are not same-resource algorithm-only gains.

Algorithm 1 Budgeted joint selection (primary sweep)

Require: Nonempty families $\{\mathcal{S}_o\}_{o=1}^{O}$, peaks $p_{o,s}$, charges $b_{o,s} \geq 0$, budget $B \geq 0$
Ensure: A primary plan π and its minimum feasible peak T^*

$E \leftarrow \text{SortByPeak}(\{(o,s) : s \in \mathcal{S}_o\})$
best$[1..O] \leftarrow \perp$; $c \leftarrow 0$; $u \leftarrow O$
if $O = 0$ **then**
  return $(\varnothing, 0)$
end if
for each equal-peak batch G_T of E, in increasing T **do**
  for all $(o,s) \in G_T$ **do**
    $h \leftarrow$ best$[o]$
    if $h = \perp$ **or** $s \prec_o h$ **then**
      if $h = \perp$ **then**
        $u \leftarrow u - 1$
      else
        $c \leftarrow c - b_{o,h}$
      end if
      best$[o] \leftarrow s$; $c \leftarrow c + b_{o,s}$
    end if
  end for
  if $u = 0$ **and** $c \leq B$ **then**
    $\pi \leftarrow \{(o, \text{best}[o]) : 1 \leq o \leq O\}$
    return (π, T)
  end if
end for
error: no feasible family (violates the no-split contract)

Algorithm 2 Private Engine with safe probe reuse

Require: Responsibility A, complete input V, query (q, r, k), completed IDs D, immutable witness metadata $\mathcal{W}$
Ensure: A local frontier F; probe results remain outside F

$I \leftarrow \text{BuildPrivateIndexes}(V)$; $H \leftarrow \varnothing$
$F \leftarrow \varnothing$; $Q \leftarrow \varnothing$ ▷ local frontier and batch
for each $a \in A \setminus D$ in the inherited order **do**
  for each $C \in \text{ReferenceCandidates}(a, I, q, r)$ **do**
    $t_\ell \leftarrow \text{LocalThreshold}(C, F, k)$
    if $\text{InheritedSafeReject}(a, C, I, t_\ell)$ **then**
      continue
    end if
    $t \leftarrow \max\{k, t_\ell, L_{\mathcal{W}}(C) + 1\}$
    $h \leftarrow \text{LookupProjectionState}(H, C)$
    if $h = \text{upper}(U)$ **and** $U < t$ **then**
      continue
    end if
    if h is not an exact state **then**
      $h \leftarrow \text{CountExactOrUpper}(C, I, t)$
      $H \leftarrow \text{StoreIfProjectionCacheable}(H, C, h)$
    end if
    if $h = \text{exact}(n)$ **and** $n \geq t$ **then**
      $(Q, F) \leftarrow \text{AppendBatch}(Q, F, (C, n))$
    end if
  end for
end for
return $\text{ReduceRemainingBatch}(Q, F)$

query coordinates of absolute value at most M, $M = 10^7$, $1 \leq r \leq 10^5$, cell width $h = r$, the local-coordinate center constructor, binary64 round-to-nearest basic operations without reassociation, correctly rounded square root, and hypot within one ulp. The implementation's accepted trimming range starts at 0.01; that wider accepted range is not the domain proved here.

Let $u = 2^{-53}$ and $\alpha = 2^{-1074}$. We use $|\,\mathrm{fl}(t) - t| \leq u|t| + \alpha$, retaining absolute terms for underflow. The common analysis allowance is $E = 10^{-4}$; it is not a runtime parameter.

1) Eligibility and accepted coverage: For exact coordinate difference Δ and its evaluated vector d, subtraction gives $\|d - \Delta\| \leq u\|\Delta\| + 2\alpha$. The evaluated squared sum S satisfies

$$S \geq (1-u)^2\|d\|^2 - 3\alpha. \tag{9}$$

Inverting these bounds, including threshold construction and squaring with relative error bounded by $4u$, yields

$$S \leq \mathrm{fl}(R^2) \implies \|\Delta\| \leq R(1+16u) + 16\sqrt{\alpha}. \tag{10}$$

One first solves the nonnegative squared-sum lower bound for $\|d\|$, then uses $\|\Delta\| \leq (\|d\| + 2\alpha)/(1-u)$ and $\sqrt{a+b} \leq \sqrt{a} + \sqrt{b}$. The factor 16 covers the small powers of $(1-u)^{-1}$ and threshold errors.

The partner test against evaluated $4r^2$ consequently gives $\|a - b\| < 2r + E$. Coverage uses the square of $\mathrm{nextDown}(\mathrm{fl}(r + \tau))$, where the stored τ differs from 0.005 by less than 10^{-18}. Since $\mathrm{nextDown}$ cannot enlarge that threshold,

$$\mathrm{cov}_r(x, z) \implies \|x - z\| < r + 0.005 + E. \tag{11}$$

The projection-substitution endpoint test is bounded by the same argument, with its own actual threshold.

2) Single-point projection: Write $D = \|a - q\|$ and let R be the computed distance. If $R \leq r$, the returned center q obeys the inverse distance bound. Otherwise, $D \leq 3M$ and

$$|R - D| \leq 8uD + 8\sqrt{\alpha}. \tag{12}$$

The computed scale is $t = \mathrm{fl}(\mathrm{fl}(R - r)/R)$, compared with $t_0 = (D - r)/D$. Use

$$\left|\frac{R-r}{R} - \frac{D-r}{D}\right| = \frac{r|R-D|}{RD} \tag{13}$$

and an absolute rounding bound for $R - r$, avoiding an assumption of well-conditioned relative subtraction. Division and subtraction contribute at most $4u + 4\alpha$ to the scale. Multiplication by the coordinate difference and final translation give a conservative center perturbation

$$160u(M + r + 1) + 64\sqrt{\alpha} < 2 \times 10^{-7} < E. \tag{14}$$

The exact projection lies at distance r from its anchor, so the actual center satisfies $\|z - a\| < r + E$. For projection substitution in a pair, the other endpoint is also checked by the reference eligibility predicate.

3) Pair centers in local coordinates: The implementation evaluates $v = \mathrm{fl}(b - a)$, $s = \mathrm{fl}(v_x^2 + v_y^2)$ and midpoint $m = \mathrm{fl}(a + v/2)$. Nonpositive s does not enter this branch; values exceeding evaluated $4r^2$ are rejected.

For normal s, nonnegative squared-sum bounds give

$$|s - \|v\|^2| \le 8u\|v\|^2 + 4\alpha, \tag{15}$$

$$\|v\|^2 \le 4r^2(1 + 16u) + 8\alpha. \tag{16}$$

Let H be the evaluated square root of the nonnegative truncation of $\mathrm{fl}(r^2) - \mathrm{fl}(s/4)$. Accounting in squared quantities, rather than dividing by a small near-diameter height, gives

$$\|v\|^2/4 + H^2 \le r^2(1 + 64u) + 16\alpha. \tag{17}$$

The squared-sum discrepancy contributes at most $10ur^2 + 4\alpha$ after scaling, the radius square at most ur^2, and subtraction and square root can be bounded by $16ur^2 + 8\alpha$. The displayed bound also covers truncation at zero.

With $d = \sqrt{s}$, $|\|v\|/d - 1| \le 16u$. Compare the evaluated perpendicular offset with $HJv/\|v\|$, where $J(v_x, v_y) = (-v_y, v_x)$. Division and coordinate multiplication give an offset error at most $64ur + 32\sqrt{\alpha}$. Orthogonality and the squared-height bound control the endpoint distance even near a diameter.

For positive subnormal s, the implementation uses $d = \mathrm{hypot}(v_x, v_y)$ instead. Positive evaluated s implies $\|v\|$ is at least of order $\sqrt{\alpha}$, so d is normal and its relative norm error is at most $4u$. Also $\|v\|^2 < 2^{-1021}$ while $r \ge 1$. The quotient H/d is below 10^{168}, excluding overflow, and absolute underflow terms are covered above.

Finally, midpoint formation, coordinate translation and endpoint subtraction yield for both endpoints

$$\begin{aligned}\|z - a\|, \|z - b\| &\le r + 256u(M + 4r + 1) + 128\sqrt{\alpha} \\ &< r + 3 \times 10^{-7} < r + E.\end{aligned} \tag{18}$$

The diameter shortcut uses a midpoint and remains subject to partner eligibility, so it obeys the same bound. Selecting either of the two candidate centers by query distance does not change their endpoint bounds.

4) Reference dependencies: For anchor a, let $Q(a)$ be centers generated by its assigned configurations and $H(a)$ their participating anchors and partners. Define

$$D_{\mathrm{ref}}(a) = H(a) \ \cup \ \{x \in P : \exists z \in Q(a), \mathrm{cov}_r(x, z)\}. \tag{19}$$

The partner, center and coverage bounds imply

$$D_{\mathrm{ref}}(A) \subseteq D(A) = \bigcup_{a \in A} \{x \in P : \|x - a\| \le 2r + 0.005 + 2E\}. \tag{20}$$

The remaining analysis establishes containment of this explicitly defined conservative set, not only observed reference accesses.

5) Floating-point cells and initial routing: Actual indices are $\lfloor \mathrm{fl}(x/r) \rfloor$. Over the stated domain they fit the integer representation. A rounded division can move a point across a nominal cell boundary only within the division error. With nominal cells defined using exact integer multiples of stored r, the point-to-cell displacement is bounded by

$$\delta_g = 8u(M + r + 1) + 8\sqrt{\alpha} < E. \tag{21}$$

This accounts for division error without assuming every computed cell has exact real-arithmetic membership.

An offset (i, j) with ideal box distance at most $2r$ has $|i|, |j| \le 3$, and the dependency enumeration radius $\lceil \mathrm{fl}(2r + \tau)/r \rceil = 3$ covers it. Its evaluated squared box distance is bounded above by $4r^2 + 256ur^2 + 64\alpha$; the evaluated threshold square is bounded below by $4r^2 + 0.019r - 64ur^2$. Since

$$320ur^2 + 64\alpha < 0.019r \qquad (r \le 10^5), \tag{22}$$

every such ideal offset is actually retained.

The ideal offset distance is

$$r\sqrt{\max(|i| - 1, 0)^2 + \max(|j| - 1, 0)^2}. \tag{23}$$

Thus an unretained offset has ideal distance at least $\sqrt{5}r$. If $a \in A_o$ and $x \in D(\{a\})$ were not routed to o, the cell errors would give $\|x - a\| \ge \sqrt{5}r - 2E$. But the dependency definition gives $\|x - a\| \le 2r + 0.005 + 2E$, implying

$$(\sqrt{5} - 2)r \le 0.005 + 4E = 0.0054, \tag{24}$$

which contradicts $r \ge 1$. Hence $D(A_o) \subseteq V_o$. Owner destination deduplication removes duplicate receivers only. The internal routing shortcut is used only when all relevant nonempty cells belong to the same continuous Hilbert owner, and shared packets preserve destination masks. These transport conditions preserve the receiver set used in the argument.

6) Physical trimming: For a full-responsibility box B_j and $x \in D(A_j)$, some $a \in A_j$ satisfies

$$\mathrm{dist}(x, B_j) \le \|x - a\| \le 2r + 0.005 + 2E. \tag{25}$$

Outward next-representable boundaries cannot increase the exact distance. Coordinate differences, maxima and hypot add at most $32u(M + 2r + 1) + 32\sqrt{\alpha} < E$, so the evaluated distance is below $2r + 0.0053$. The actual outward-rounded threshold exceeds $2r + 0.019999999$, and accepts the record. Unsafe trimming retains the input; empty responsibility has empty dependencies. Combining both levels proves

$$D_{\mathrm{ref}}(A_j) \subseteq D(A_j) \subseteq V_j \subseteq V_o. \tag{26}$$

The largest displayed projection, pair-center, cell and trimming error bounds are respectively below 1.80×10^{-7}, 2.96×10^{-7},

8.98×10^{-9} and 3.63×10^{-8}. Finite regression checks accompany this derivation; they are not its mathematical premise. Java's documented $\mathrm{sqrt/hypot}$ guarantees are the assumed function specifications (https://docs.oracle.com/javase/8/docs/api/java/lang/Math.html).

B. Auxiliary Result-Preservation Arguments

1) Reference semantics and execution requirements: Fix the reference finite configurations, common numerical predicates and responsibility assignment from main-paper Section II. A completed probe anchor has finished its assigned configurations using only frontier-safe pruning. Parent replays used for cost measurement do not receive additional completion credit; the decomposition below concerns credited initiation responsibilities rather than absence of repeated measurement work. Every transferred witness has a valid complete count or conservative coverage lower bound under these same predicates.

The contract requires common construction and coverage predicates, conservative numerical dependency envelopes, complete task inputs, and pruning that preserves every non-dominated reference objective. In particular, an anchor-count upper bound is usable only when its counted region is certified to contain the candidate's accepted records.

Lemma 1 (Dependency closure)**.** *If $D(A_{o,j}) \subseteq V_{o,j}$, every configuration assigned to task j has its required generating partner and all accepted counting records in that task's input.*

Proof. For an anchor a and candidate center z, the numerical envelope bounds every accepted record by $\|x - a\| \le 2r + 0.005 + 2e$ and also contains the generating partner. Both belong to $D(A_{o,j})$, hence to $V_{o,j}$. The task envelope in the main-paper private-input equation must preserve this inclusion. □

Lemma 2 (Probe–formal decomposition)**.** *Let original tasks partition A_o and let P_o contain probe-completed anchors. Probe initiation together with formal sets $R_{o,j} = A_{o,j} \setminus P_o$ covers each reference generating configuration exactly once as an initiation responsibility.*

Proof. Each configuration has one responsible anchor. That anchor belongs to one owner and one original task. It either belongs to P_o and was completed by the probe, or belongs to one remaining set $R_{o,j}$. The cases are disjoint and exhaustive. Thus, with $G(a)$ denoting assigned configurations,

$$\bigcup_{a \in A_o} G(a) = \bigcup_{a \in P_o} G(a) \cup \bigcup_{j} \bigcup_{a \in R_{o,j}} G(a). \tag{27}$$

Visible points are retained in both cases. The statement assigns configurations uniquely even when different configurations describe the same circle. □

Lemma 3 (Admissible witness threshold)**.** *Let a retained feasible probe circle S have valid count lower bound n_S. A formal candidate C with $d(S) < d(C)$ and complete count at most n_S is dominated by S. Consequently $L(C) + 1$ is a safe necessary accepting count.*

Proof. The actual count of S is at least n_S, so it is no smaller than that of C, while its distance is strictly better. It therefore dominates C. Maximizing the valid lower bounds over strictly distance-better witnesses preserves this implication. If a partial count c and conservative remaining upper bound u satisfy $c + u < L(C) + 1$, C cannot meet the necessary count and may be rejected. A rejected partial count is not stored as an exact objective. □

Theorem 1 (Objective-frontier preservation)**.** *For an execution satisfying the reference and numerical contract above, with disjoint task responsibilities, complete task inputs, retained probe results and admissible local pruning, DiSMC returns the reference non-dominated objective frontier. If export additionally satisfies the main-paper export-inclusion condition and uses the common predicate, each retained representative has its complete query-input members, yielding an equivalent answer under the stated output contract.*

Proof. Lemma 2 assigns every reference configuration to a probe or a formal task. Lemma 1 preserves its generation and complete count dependencies. The execution contract preserves safety of the inherited pruning rules, and Lemma 3 makes transferred thresholds another valid rejection rule. Every pruned objective is dominated by a retained feasible witness or its dominating successor. A finite dominance chain ends in a non-dominated objective. Finally,

$$\mathrm{Sky}\left(\bigcup_t C_t\right) = \mathrm{Sky}\left(\bigcup_t \mathrm{Sky}(C_t)\right) \tag{28}$$

for complete objective sets. Merging all formal frontiers with the retained probe frontier therefore preserves the reference objective frontier. Each retained circle is a reference candidate with a complete count. The main-paper export-inclusion condition ensures every accepted source record reaches its representative, and the common predicate excludes nonmembers. Exported members therefore realize that count and objective. Together with frontier equality, this proves the stated answer equivalence. □

2) Export condition and proof scope: The main-paper export-inclusion condition is separate from task-input completeness. Under it, every accepted member reaches its circle's predicate test, and that test writes no nonmember; hence $M(C) = \{x \in P : \mathrm{cov}_r(x, z_C)\}$. The export-envelope analysis below establishes this condition for certified experimental radii and the stated finite/index domain; other radii require the same arithmetic certificate. Main-paper Section III's two-level proposition does not imply it. Recorded member-count checks are finite implementation evidence, not a substitute for that export-index proof.

3) What one representative preserves: Reference circles with equal evaluated distance and cardinality form an objective-equivalence class. Exporting one circle and its complete source members per non-dominated class preserves the available objective trade-offs, not every tied center or group.

Applications requiring all tied groups need an extended output contract; Theorem 1 covers representative answers.

Scope of the evidence. ESMC supplies the finite reduction. Lemmas 1–3 and Theorem 1 establish the distribution and transfer guarantees under the stated numerical inclusion and pruning conditions. Implementation tests check these conditions on finite cases; they do not prove continuous completeness or universal floating-point correctness.

4) Distinct completion and visibility states: The interface maintains (R, V, F): remaining initiation, dependency visibility and retained feasible results. Completion updates R; input planning changes V subject to closure; witness transfer reads F. Removing a completed anchor from V can destroy another anchor's generating partner or covered member. Discarding its frontier from F can lose a non-dominated objective. Typed certificate reuse additionally obeys the main-paper upper-certificate proposition.

5) Numerical domain, representatives and fallback: The implementation permits trimming for finite coordinates of absolute value at most 10^7 and radii in $[0.01, 10^5]$, retaining original inputs outside that domain. The main-paper two-level input proposition establishes the two input inclusions for the narrower stated nonnegative-source domain with $r \geq 1$, $\beta = 1$ and the local constructor; it does not assert coverage of every accepted parameter. The supplementary derivation accounts for the 0.02 margin and outward rounding. Anchor-count rejection separately uses an actual-center inclusion certificate and is skipped when that certificate fails.

Local and driver reductions choose smaller center X, then Y, among surviving tied rounded objectives. This makes a retained set deterministic. Global canonical identity additionally requires every tie-pruning step to preserve the minimum common representative key; objective-frontier preservation alone does not provide this stronger contract. Continuous optimality uses the inherited reduction.

C. Complete Member Export: Envelope and Cell Inclusion

This argument concerns the implemented export lookup, not task routing. Assume finite binary64 round-to-nearest coordinates and arithmetic, positive cell width h, and finite cell-index ranges within the integer representation. Let $a = \mathrm{fl}(r + 0.005)$, $t = \mathrm{nextDown}(a)$ and $T = \mathrm{fl}(t^2)$. The exporter uses $E = \mathrm{nextUp}(a)$ and coordinate endpoints

$$\ell = \mathrm{nextDown}(\mathrm{fl}(z - E)), \quad b = \mathrm{nextUp}(\mathrm{fl}(z + E)). \tag{29}$$

It inserts the circle into every cell from $\lfloor \mathrm{fl}(\ell/h) \rfloor$ to $\lfloor \mathrm{fl}(b/h) \rfloor$ on each axis. The source uses the same positive h and evaluated division for its cell indices.

1) A checkable radius certificate: Let d_* be the largest nonnegative binary64 value satisfying $\mathrm{fl}(d_*^2) \leq T$, and $d_+ = \mathrm{nextUp}(d_*)$. Define the exact rational midpoint $B = (d_* + d_+)/2$. The sufficient certificate is

$$\mathrm{fl}(d_*^2) \leq T < \mathrm{fl}(d_+^2), \qquad B < E. \tag{30}$$

This is a parameter-specific arithmetic certificate, not a sampled-point correctness test. The released script finds adjacent d_*, d_+ around $\sqrt{T}$, checks their rounded products with exact rational arithmetic, and compares B and E as rationals. Certificates are released for the main online/full-data radii 20, 40, 60, 75, 100, 150, 200 and 300 and the listed sensitivity radii. Arbitrary additional radii require the same certificate; no unimplemented runtime fallback is assumed.

2) Accepted points are inside the envelope: Write $d_x = \mathrm{fl}(z_x - x)$ and $d_y = \mathrm{fl}(z_y - y)$. A point is accepted when

$$\mathrm{fl}(\mathrm{fl}(d_x^2) + \mathrm{fl}(d_y^2)) \leq T. \tag{31}$$

Nonnegative addition and monotonic round-to-nearest imply each rounded square is at most T, hence $|d_x|, |d_y| \leq d_*$. The rounding preimage of values no larger than d_* has magnitude at most the midpoint B (including a possible tie). Therefore the exact stored-coordinate differences satisfy $|z_x - x|, |z_y - y| \leq B < E$. Each accepted coordinate lies strictly inside the exact interval $[z - E, z + E]$. Adjacent outward rounding of its computed endpoints contains this real interval, so $\ell \leq x \leq b$ on both axes.

3) Envelope coordinates map to included cells: Division by the same positive h, correctly rounded division and floor are monotone. Thus $\ell \leq x \leq b$ implies

$$\lfloor \mathrm{fl}(\ell/h) \rfloor \leq \lfloor \mathrm{fl}(x/h) \rfloor \leq \lfloor \mathrm{fl}(b/h) \rfloor. \tag{32}$$

The exporter enumerates this inclusive integer range; consequently every accepted point's cell contains the representative in its lookup list. This proves the export-inclusion condition for certified radii and the stated finite/index conditions, including points arbitrarily close to cell boundaries. It does not rely on exact real-arithmetic membership of the nominal grid cell. Applying the same predicate at export then admits all and only the required members. Stable one-pass point traversal and unique insertion per circle/cell prevent duplicate output for a given source record and representative.

The scope is the current partitioned export path and common numerical predicate. The result is distinct from finite member-count checks and does not claim that every parameter accepted by the application has been certified. Source and query bounds from the main numerical domain bound generated centers; the actual prepared coordinates and index range must satisfy the finite/int assumptions. The certificate does not prove the absence of I/O failures; the reported completed runs validate successful writing and count aggregation.

VII. Exact Values Behind Main-Text Figures

The following tables retain exact aggregates used by the main-text and supplementary online-query, single-machine, input-policy and probe-reuse figures. Values are unchanged from the corresponding manuscript records. Means, sample standard deviations, medians and timeout censoring retain their original definitions; no new experiment or statistical estimate is introduced. The original per-run records and detailed protocols remain in the preceding sections.

References

[1] Microsoft Research, "Geolife GPS trajectory dataset—User Guide," 2011. [Online]. Available: https://www.microsoft.com/en-us/research/publication/geolife-gps-trajectory-dataset-user-guide/.

TABLE XXVIII
Fixed queries: complete online E2E mean ± sample SD (s), three runs per arm; 3M points, 40 cores, 5% cap. GeoLife $r = 100$; TDrive $r = 300$.

Data	Query	DiSMC	Static-Greedy	Reduction
GeoLife	q20	1420.252 ± 72.362	2884.073 ± 507.864	50.76%
GeoLife	q40	1312.631 ± 83.379	3232.610 ± 187.563	59.39%
GeoLife	q60	1000.158 ± 141.715	2544.110 ± 165.397	60.69%
GeoLife	q80	1190.935 ± 125.618	1846.305 ± 261.438	35.50%
TDrive	q20	511.974 ± 10.510	859.738 ± 10.797	40.45%
TDrive	q40	408.719 ± 27.636	822.669 ± 24.273	50.32%
TDrive	q60	422.884 ± 45.541	856.135 ± 25.956	50.61%
TDrive	q80	518.346 ± 41.508	793.458 ± 21.848	34.67%

TABLE XXIX
Single-threaded ESMC versus 40-core DiSMC: complete size series, three-run median seconds. Timeout: the 7200-second limit was reached.

	GeoLife		TDrive		Porto	
Size	ESMC	DiSMC	ESMC	DiSMC	ESMC	DiSMC
100K	22.261	22.969	20.205	20.194	15.346	20.019
200K	115.590	24.801	103.165	21.586	79.861	20.723
400K	1271.005	28.901	862.560	25.750	1026.332	22.115
600K	Timeout	33.927	4933.331	27.915	5254.248	23.084
800K	Timeout	38.153	Timeout	27.605	Timeout	23.887
1M	Timeout	43.433	Timeout	31.695	Timeout	25.323

TABLE XXX
Full/Dependency input-planning control: three-run median E2E (seconds), fixed acquisition before calibration, common work signal and 5% cap.

Data	r/r_0	Full	Dep.	Reduction
GeoLife	0.75	76.937	67.976	11.65%
GeoLife	1	110.397	104.668	5.19%
GeoLife	1.5	226.947	193.276	14.84%
TDrive	0.75	64.548	59.374	8.02%
TDrive	1	79.650	72.788	8.62%
TDrive	1.5	200.398	172.427	13.96%

TABLE XXXI
Completion reuse (R) and witness transfer (W): three-run mean E2E ± sample SD in seconds. All four configurations use identical responsibilities, inputs and probe IDs; only the two transfer switches vary. Checks are task-summed formal operations, unchanged across recorded repetitions.

		E2E (s)		Checks (10^9)	
R	W	Geo	TDrive	Geo	TDrive
0	0	103.979 ± 1.613	82.546 ± 2.297	140.804	112.340
0	1	96.355 ± 0.703	73.637 ± 1.097	71.026	33.446
1	0	100.971 ± 1.342	77.870 ± 0.752	140.121	112.171
1	1	93.622 ± 0.902	69.876 ± 1.257	70.503	33.332

[2] J. Yuan et al., "T-drive: driving directions based on taxi trajectories," Proc. ACM SIGSPATIAL GIS, 2010, pp. 99–108.

[3] UCI Machine Learning Repository, "Taxi Service Trajectory—Prediction Challenge, ECML PKDD 2015," 2015, doi: 10.24432/C55W25.

[4] M. Tang, Y. Yu, A. R. Mahmood, Q. M. Malluhi, M. Ouzzani, and W. G. Aref, "LocationSpark: In-memory Distributed Spatial Query Processing and Optimization," Frontiers in Big Data, vol. 3, article 30, 2020, doi: 10.3389/fdata.2020.00030.

[5] A. M. Aly et al., "AQWA: Adaptive Query-Workload-Aware Partitioning of Big Spatial Data," Proc. VLDB Endowment, vol. 8, no. 13, pp. 2062–2073, 2015, doi: 10.14778/2831360.2831361.